Banana Case Study

A study of banana retail prices in the United Kingdom

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1. Abstract

Background: Access to fresh produce is a major determinant of economic health of a population. Bananas are a good source of potassium and other beneficial micro-nutrients and are a staple fresh fruit of many households. The trends in banana prices have seen a sharp increase in recent years, and are considered as a proxy for the health of the global trade economy. The study aimed to highlight the effect of region of origin on the price of bananas in the United Kingodom, when adjusting for year and time of year between 2020 and 2025.
Methods: The study used four supervised learning techniques to model banana prices in the United Kingdom between the years 2020 and 2025, namely: multiple linear regression, single deep regression tree, AdaBoost tree and Random Forest. A training set and validation set were used to fit and tune models, and a final independent test set was used to evaluate their performance.
Results: The best performing model was a Random Forest which explained 71% of the variance in the retail price of bananas in the United Kingdom (UK). None of the variables were found to be strong determinants of banana price in the Random Forest model. Cost Rica and Ghana were found to be the most important regions of origin in predicting banana prices by the Random Forest model. The multiple regression model found that banana prices were lower when sourced from Ghana and higher when sourced from Costa Rica, as compared to those sourced from the European Union (EU).
Conclusion: The price of bananas is lowest during the fourth quarter of the year, in the cold months and highest during the second quarter, during the warm months. When accounting for time of year and year, all the models had fairly good fits to the data but Random Forest was the clear best performing model and only model to explain more than 70% of the variance in the data. Bananas sourced from Africa had the lowest retail prices whereas bananas sourced from South America had the highest retail prices, as compared to bananas sourced from the EU.

2. Introduction

Bananas are common fruits in many Western households. They are grown in tropical climates, closer to the equator. The global economy has changed considerably since 2020 due to the Covid-19 pandemic. Furthermore, there has been increased strain on supply chains in recent years with conflict in Gaza and Ukraine as well as trade tariffs. Given the sharp increase in banana price since 2020, it is crucial that the factors associated with banana price be identified and evaluated.

With limited available research quantifying the effects of region of origin when adjusted for year and month, this study aimed to use supervised learning models to evaluate effect of region of origin on banana price.

3. Methodology

The following section describes the data collection as well as statistical techniques employed in this study.

3.1 Data collection

The data used in this study was obtained from the United Kingdom Governmental website. It was collected by the United Kingdom Department for Environment, Food & Rural Affairs fortnightly between the years 2015 and 2025 (***United Kingdom Department for Environment, Food & Rural Affairs, no date***). The prices are national averages of the most usual prices charged for bananas at wholesale markets in Birmingham and London. The data collected from the markets were assumed to be independently and randomly sampled and were scoped to just consider prices between 2020 and 2025.

3.2 Measurement

This data gives the average wholesale prices of bananas by country of origin, the country of origin and the day, month, year collected. All prices are in pounds (£) per kg. 17 unique region of origin were considered for each price including the European Union (EU). The dataset was scoped down to consider just the last 5 years of banana price.

3.3 Statistical techniques

Multiple linear regression modeled the linear relationship between predictors (origin, year and month) and the average banana price, expressing this relationship in a single linear equation with estimated coefficients. Tree models used binary partitioning of the predictor space to model the relationships between predictors and the average banana price. Single deep decisions trees form a long chain of conditionals whereas AdaBoost adaptively fits multiple shallower trees in parallel. Where the previous models may unintentionally focus on a single predictor, Random Forests grows multiple trees by randomly selecting a predictor to prevent overfitting. The absolute magnitude of the multiple linear regression coefficients expresses their relative importance in the model and their sign indicates the direction of the relationship with average banana price. Feature importance in a tree expresses the prediction power of a predictor by the proportion of nodes determined by the predictor.

4. Results

The following section reports the findings of the exploratory data analysis as well as the results of the four supervised learning models.

4.1 Banana profile

The price of bananas since 1995 has experienced a steady rise followed by sharp leap of almost 50% in banana prices in the last five years (**Figure 1**). This sharp increase may be caused by the global recession and strained supply chains during and following the Covid-19 pandemic and wars in Ukraine and Gaza. The average banana price per kilogram (kg) differs slightly by time of year, where prices are lower in colder months and higher in warmer months (**Figure 2**). The overall distribution of banana price since 1995 (**Figure 3**) is slightly right skewed due to an anomaly in Nicaragua in 2013. The distribution of banana price by month (**Figure 4**) differs considerably over the course of a year hence, the time of year may be a strong predictor of banana price. The distribution of banana price by origin also differs considerably (**Figure 5**;**Figure 6**) hence, the region of origin may be a strong predictor of banana price.

4.2 Model evaluation

As indicated in the comparison of model performance (**Figure 7**), all the models had fairly good fits to the data however Random Forest had the higher R^2 of the four models. The mean square error (MSE) and mean absolute error were consistent across all models, implying that there were were no clear outliers affecting model performance.

4.3 Feature importance

The following section discusses the importance of region of origin, year and month in the Random Forest model and the multiple linear regression model.

4.3.1 Effect of year on banana price

The importance of each year in predicting banana price could be indicative of the difference in rate of change in banana price relative to 2020 (**Figure 8**). Where the banana price between 2020 and 2021 experienced the greatest rate of change, the prices naturally experiencing some stabilisation in 2022 and 2023 and then fluctuating again in 2024. The multiple linear regression model coefficients being larger and positive in recent years may be indicative of the overall change in banana price from 2020 to 2025 hence, 2025 having being the strongest determinant of banana price (**Figure 9**).

4.3.2 Effect of time of year on banana price

March, April and May had the highest average banana price whereas October, November and December had the lowest (**Figure 2**). This aligns with the Spring and Autumn months of the Northern hemisphere, respectively. The greatest rates of change in banana price are reflected in the relative importance of these months in the Random Forest and the direction and magnitude of the coefficients in the multiple linear regression model (**Figure 8**;**Figure 9**).

4.3.3 Effect of region of origin on banana price

Costa Rica and Ghana were the most important regions in predicting banana price in the Random Forest however, all features had a score of less then 0.2 which implies that none of the features were particularly strong predictors of banana price (**Figure 8**). The multiple linear regression model found Ghana to be the strongest predictor of banana price (**Figure 9**). As indicated by the absolute coefficient magnitude, the multiple linear regression model found banana price to be lowest when sourced from Ghana as compared to the EU and highest when sourced from Brazil (**Figure 9**).

5. Discussion and Conclusion

Access to fresh produce is a canary in the coal mine of economic healthy of a region. While seasonal changes in banana prices remain consistent, the large differential in the banana prices from 2020 to 2025 in the UK may be indicative of the future of economic health of the region in years to come. Global markets are still recovering from the Covid-19 pandemic and strains on supply chains from political instability represent possible mechanisms for the lower price of bananas sourced from Africa as compared to the prices of bananas sourced from South America.

The difference in the performance of the Random Forest model compared to the other tree models may be indicative of the other tree models over-reliance on the strong predictors of years and months. The difference in the performance of the Random Forest model compared to the multiple linear regression model may be indicative of the need to model the interactions between years, months and regions of origin.

Future research should consider further study of other fresh produce to identify if the trends in banana price are present in other fresh produce as well as contrast the effects of predictors on banana price in previous years to current effects.

Consumers should ready themselves for further increases in prices and be aware of the country of origin when comparing banana prices. Policy makers should consider mitigation strategies, in light of the trends in fresh produce access.

6. References

United Kingdom Department for Environment, Food & Rural Affairs (no date) Machine-readable: Wholesale banana prices weekly 2015 to 2025. ***https://www.gov.uk/government/statistical-data-sets/banana-prices***.

7. Addendum A: Figures

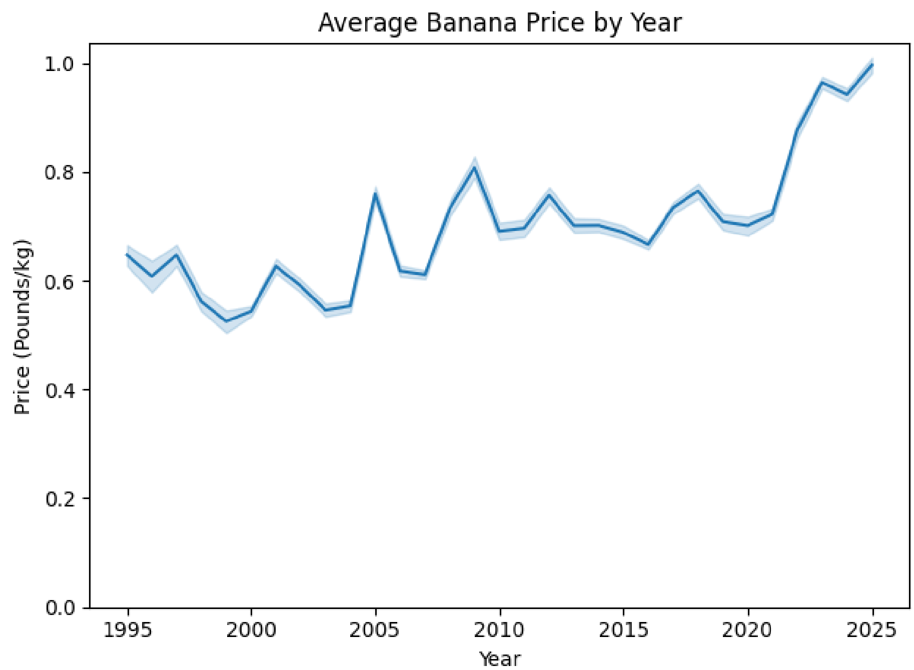

Figure 1: Average Banana Price by Year

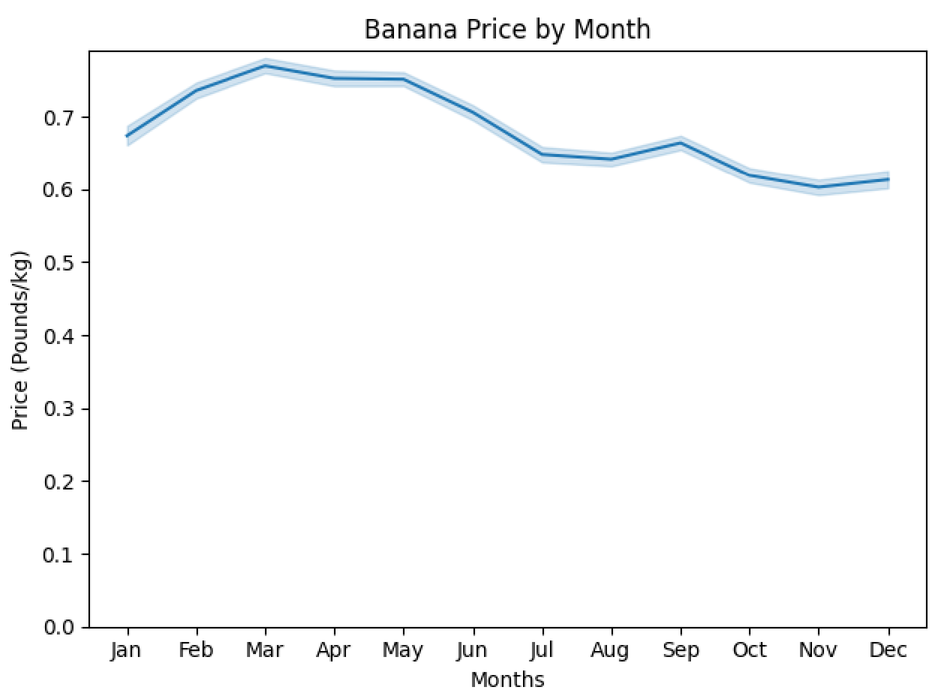

Figure 2: Average Banana Price by Month

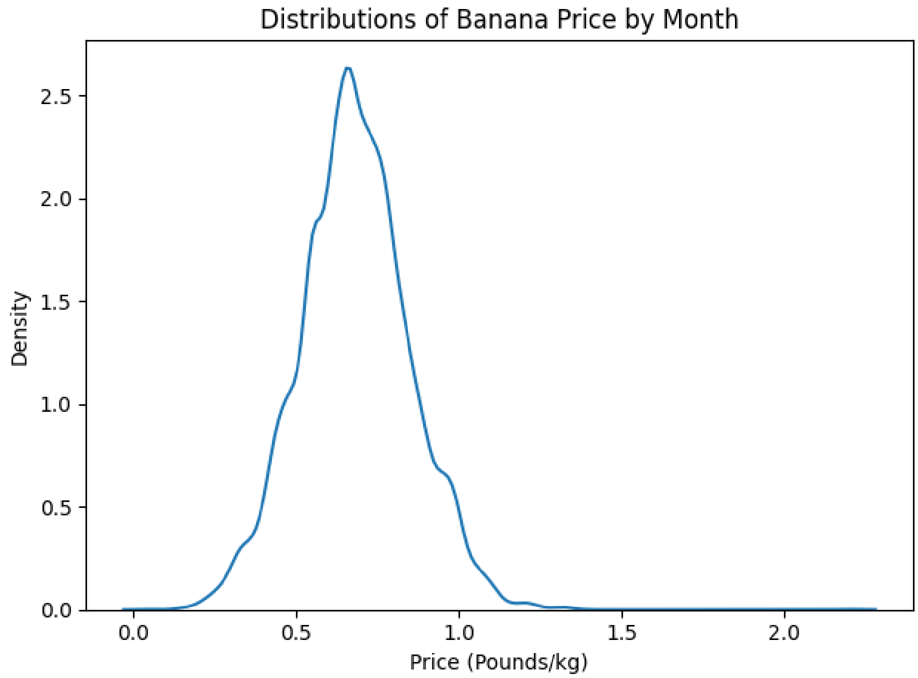

Figure 3:Distribution of Average Banana Price

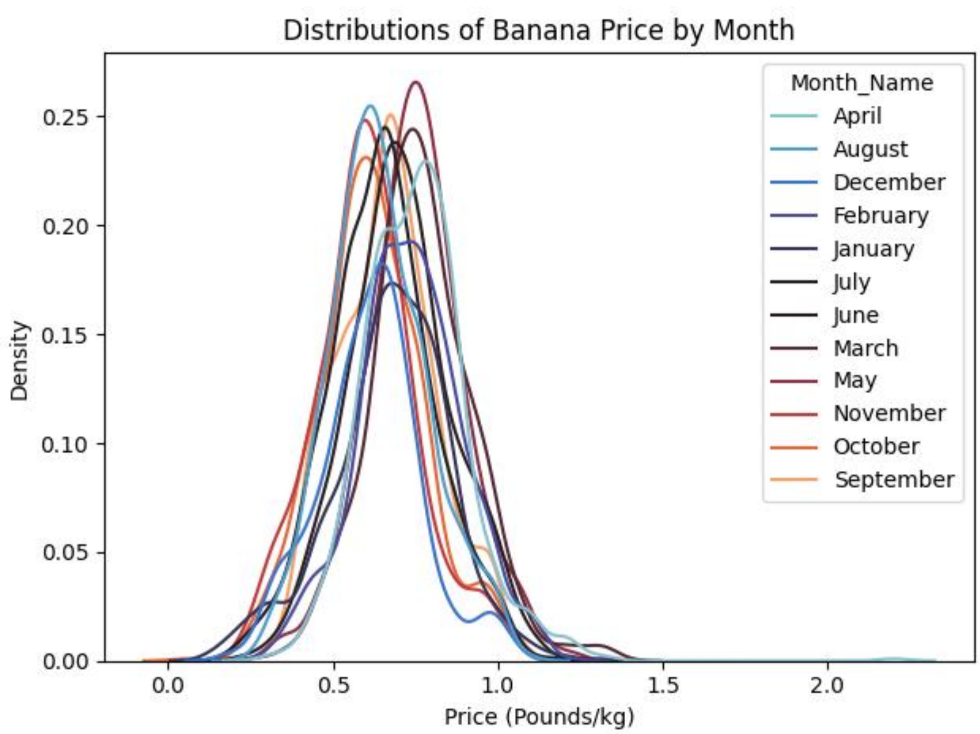

Figure 4: Distribution of Banana Price by Month

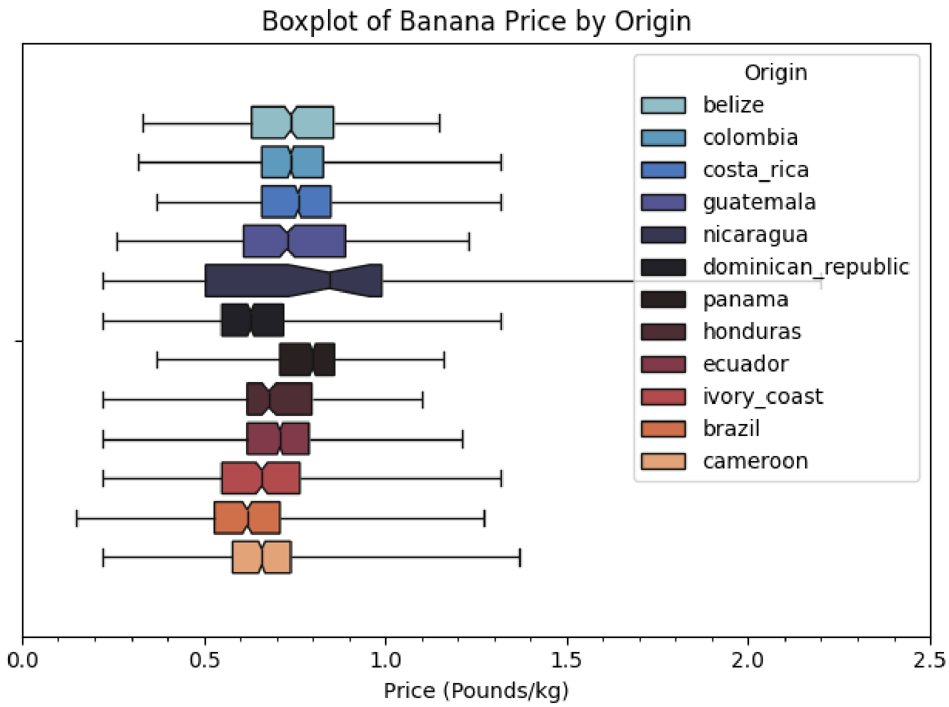

Figure 5: Banana Price by Origin

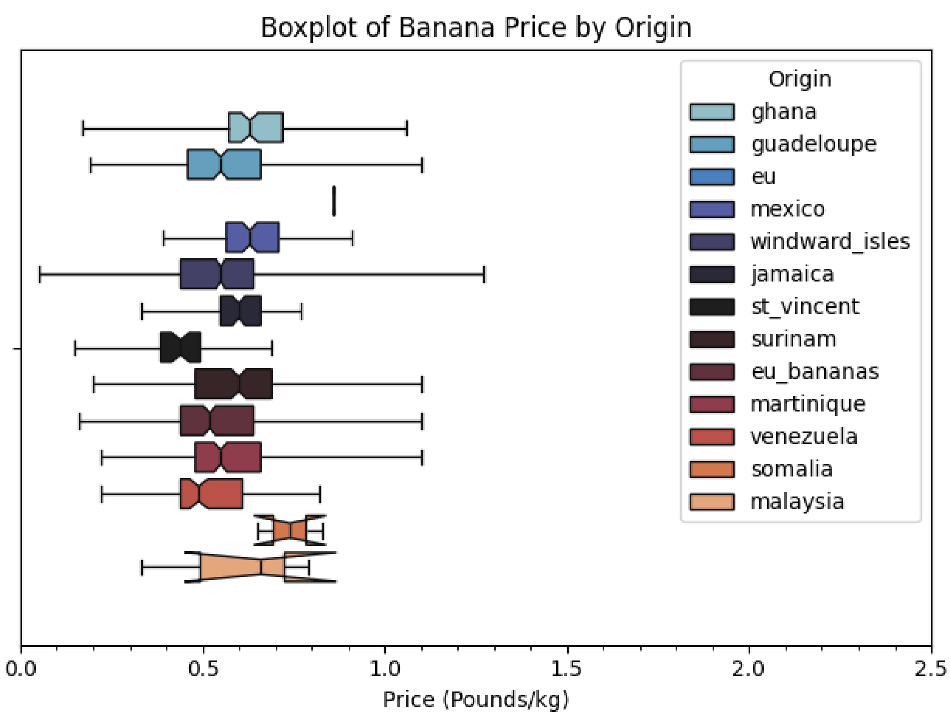

Figure 6: Banana Price by Origin

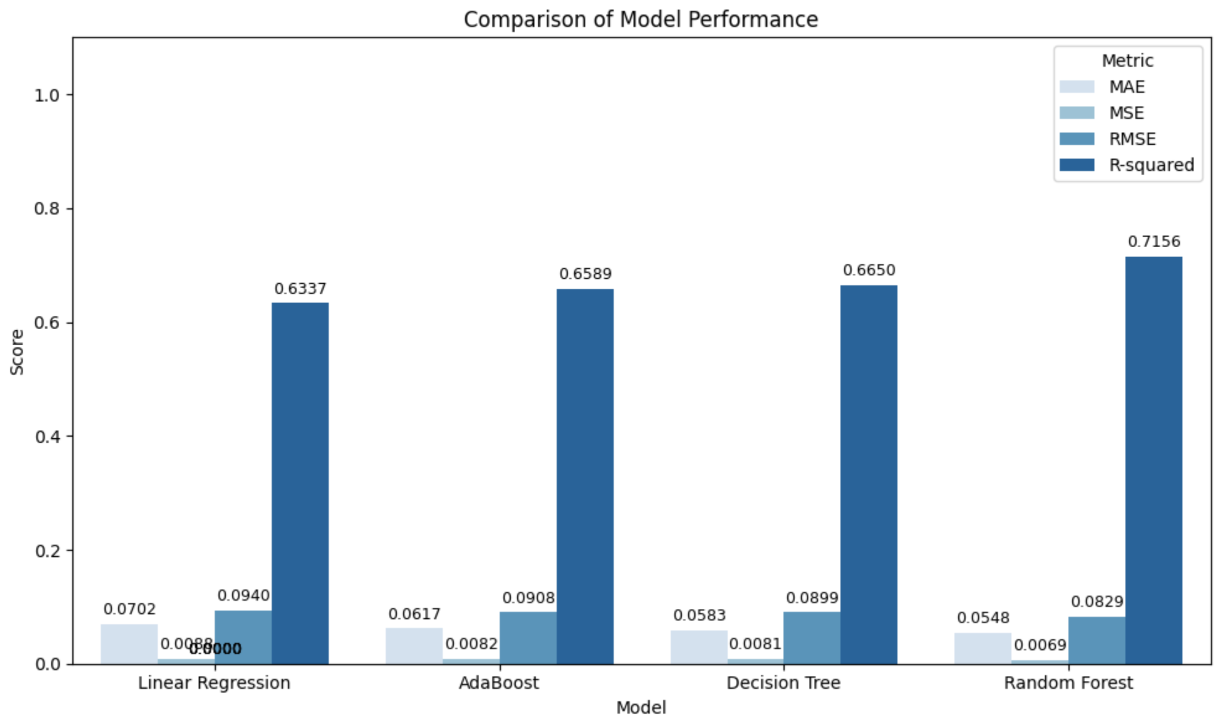

Figure 7: Comparison of Model Performance

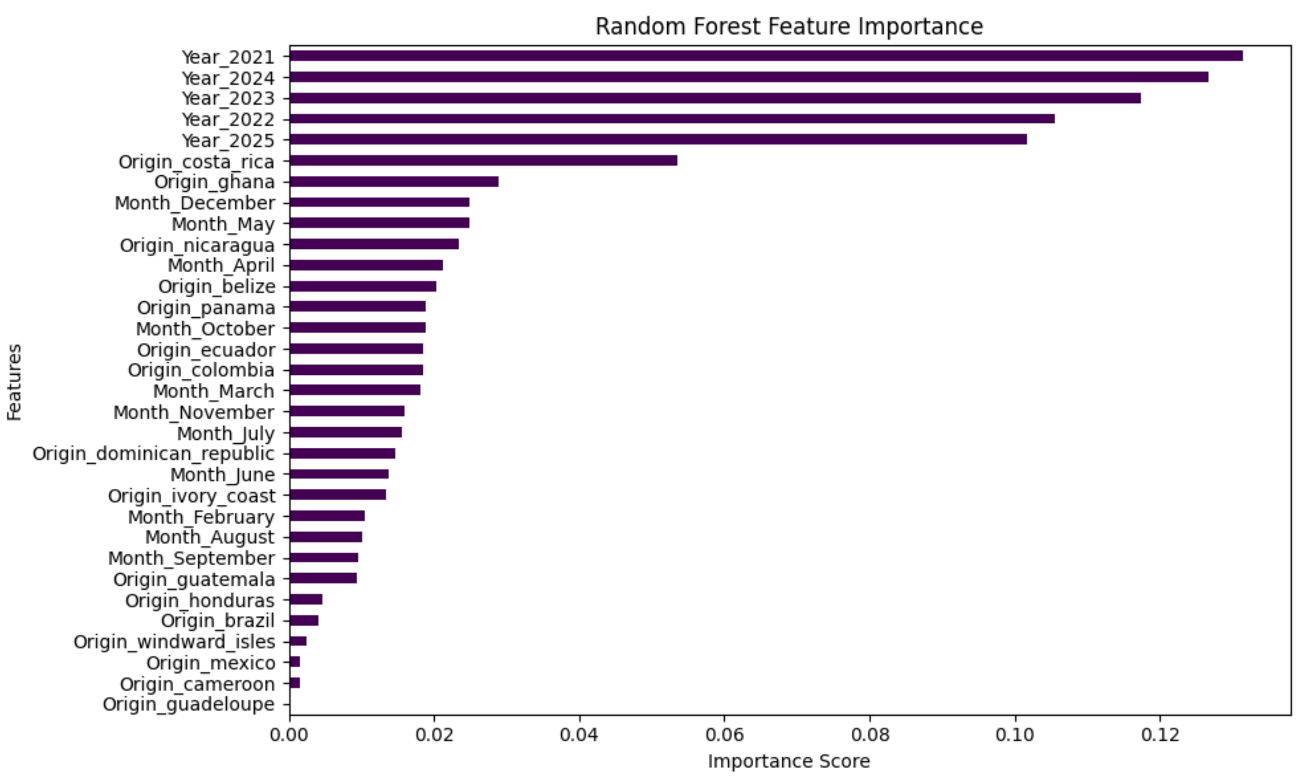

Figure 8: Random Forest Feature Importance

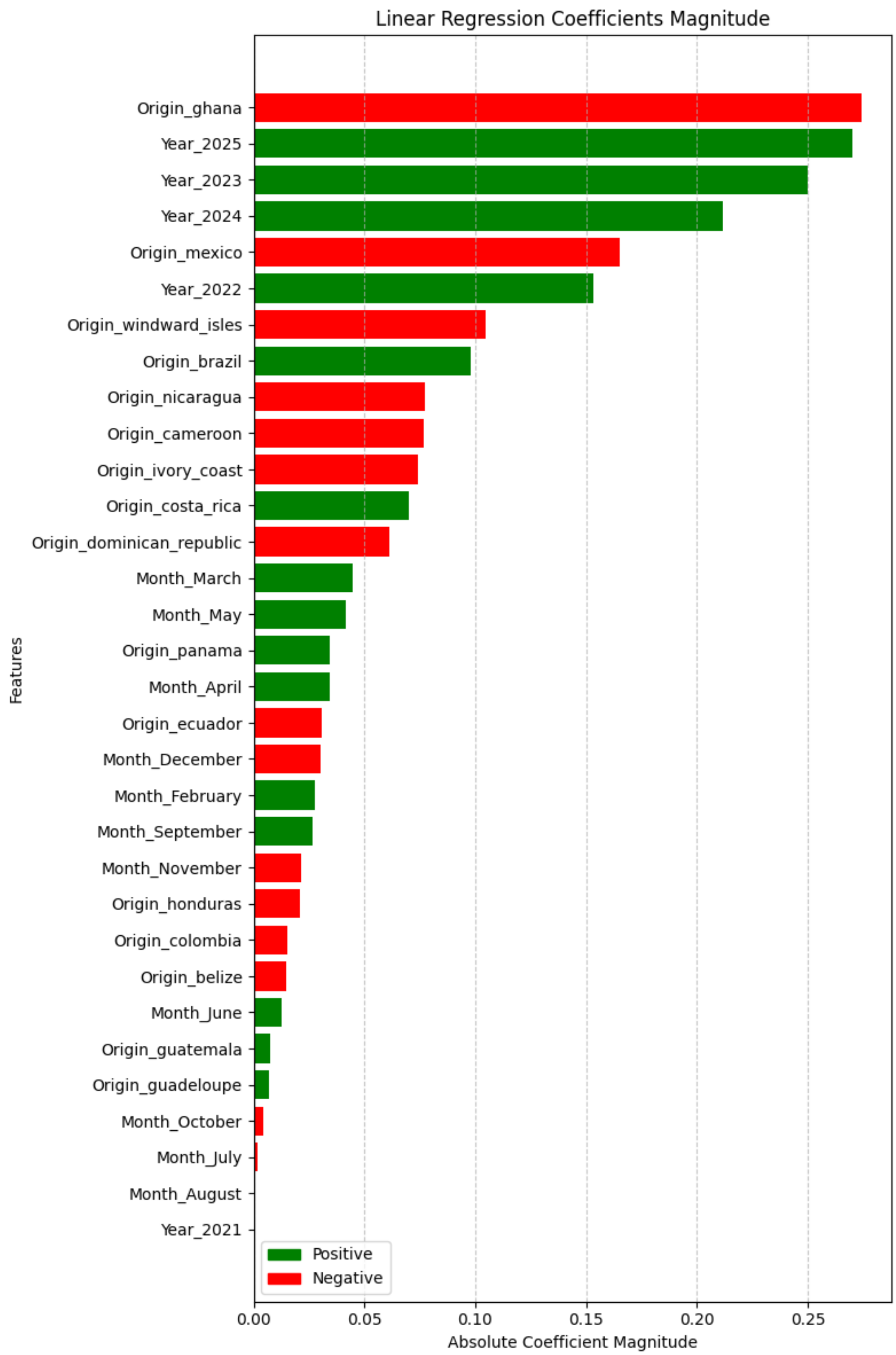

Figure 9: Multiple Linear Regression Feature Importance